

Yashwanth Muppidi

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EDUCATION

Purdue University

Bachelors of Science in Mechanical Engineering

Expected Graduation: May 2027

GPA: 3.6

TECHNICAL EXPERIENCE

Zipline | Mechanical Engineering Intern

01/2026 - 08/2026

- Developed a portable UAV data offloader, co-designing a custom PCBA to interface with the UAV, NUC, microcontroller, modular batteries, UI, cooling system, and COTS components with mounting hardware
- Optimized the data offloader for DFM/DFA and ~50 unit production, using low-cost MJF and standard COTS parts
- Engineered a rail-guided, ratcheting VTOL drone transport fixture in NX using stress analysis to validate 80/20 and sheet metal structure to reduce operators by 50% and field operation time by ~80% in commercial deployment
- Deployed RTK GNSS hardware for structural stability testing, developing a Python tool to process position data
- Built an end-of-line hardware test enclosure with machined, laser-cut, and 3D-printed components, integrating DIN rails, custom fixturing, and electrical interfaces to validate PCBA, RF, and electromechanical subsystems
- Integrated a Teensy-based RF test system using IIR filtering for automated pass/fail outputs with adjustable thresholds
- Designed test fixtures to conduct thermal characterization and highly accelerated life testing of charging dock systems

Johnson & Johnson MedTech | R&D Product Development Engineering Intern

05/2025 - 08/2025

- Prototyped a surgical robotics end-effector, leveraging resin 3D printing to rapidly iterate and integrate poka-yoke features to resolve tolerance stack-ups and component fit issues
- Developed an embedded control stack on an Arduino Mega, using a CNC shield to drive stepper motors and a Nintendo Nunchuck mapping I2C inputs to 3-DOF motion
- Designed ~10 fixtures to significantly reduce components' assembly time using machined 80/20 and SLA 3D prints
- Led root-cause analysis on a product-formulation test machine by conducting structured tests to diagnose motor stalling and distance sensor failures, then documenting and implementing corrective actions
- Analyzed DFM/DFA for injection-molded plastics and sub-assemblies of medical devices, updated SolidWorks drawings for manufacturing, and performed performance validation tests to verify design specifications

NEMO | Humanoid Robotics Purdue Trace Labs Researcher

01/2025 - Present

- Co-designed several iterations of a 20-DOF humanoid robot (12 DOF in the legs, 8 DOF in the upper body) using SolidWorks, incorporating modular components to streamline design and simplify manufacturing
- Co-developed direct-drive mechanisms with dual motor configurations for the hip, knee, and ankle pitch joints optimizing torque efficiency while reducing mechanical complexity
- Leveraged ANSYS FEA & hand calculations to analyze deformation, stress distribution, and joint load-bearing capacity, reducing weight by ~50% by optimizing material thickness and selection without compromising structural integrity
- Fabricated prototypes out of aluminum and PLA, mounting the final assembly onto a custom 80/20 test stand
- Analyzed MuJoCo physics simulations to assess gait stability and joint loads to improve mechanical performance

DuPont | Mechanical Engineering Intern

05/2024 - 08/2024

- Co-engineered a mechanical lifting fixture designed to support a 1200-pound load, increasing its strength by 10x
- Analyzed fixture stress distribution with the lead engineer and equipment specialist, reducing weight by 30%

Purdue Humanoid Robotics Club | Head of Engineering

04/2024 - Present

- Directed 250+ club members across sub-teams to design humanoid, quadruped, teleoperated arm, & hand prototypes
- Communicated with and presented to corporate sponsors, showcasing technical progress to help secure over \$300K in funding and essential hardware
- Coordinated communication between mechanical, electrical, and software team leads, executives, and professors to facilitate adherence to project timelines and enhance progress toward the organization's objectives

Purdue Aerial Robotics Team | R&D Testing Engineer

08/2023 - 05/2024

- Improved VTOL drone's fuselage & empennage structural design for reliable vertical-to-horizontal flight transitions
- Designed a SolidWorks prototype mount for V2 fuselage wind tunnel tests, optimizing aerodynamic performance
- Co-developed a custom test rig with an integrated load cell to measure motor thrust and carbon fiber spar deflection

SKILLS

Engineering: SolidWorks, NX, Ansys FEA, DFM/DFA, GD&T, 3D Printing, Machining, Laser Cutting, PCB Design

Programming: MATLAB, Python, C, Arduino IDE, Git, MuJoCo, ROS 2 | **Productivity:** MS Office, Google Suite, Jira